Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions required for a system to be in equilibrium.

1 .....

.....

2 .....

..... [2]

(b) Figure 1.1 shows a box hanging from a horizontal uniform beam of length 2.0 m. The beam is attached to the wall by a hinge and a cable at point P. Point M is the centre of mass of the beam.

The box has a mass of 4.0 kg, and the beam has a mass of 2.0 kg.

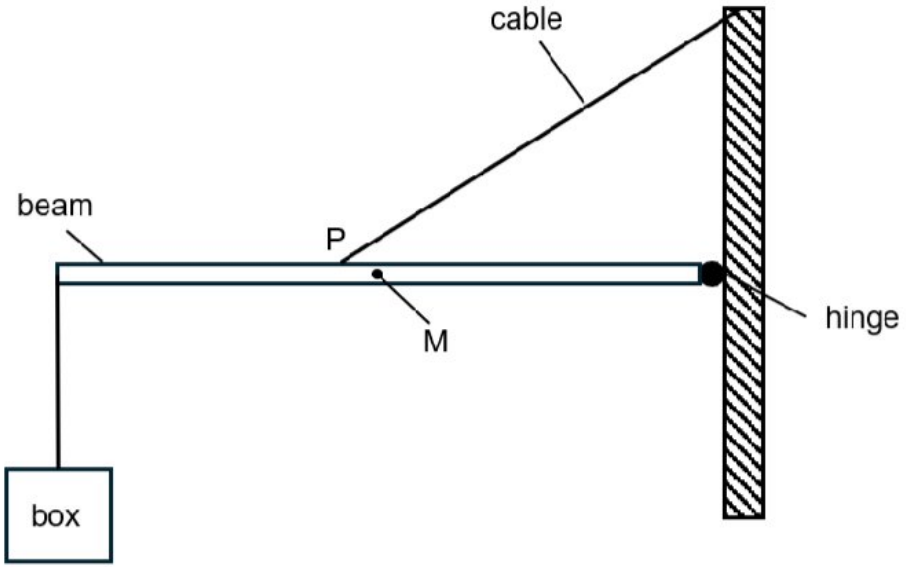


Figure 1.1

(i) On Figure 1.1, draw

1. the possible line of action of the combined weight of the beam and the box. Label the line L. [1]

2. the force the hinge exerts on the beam. Label the force *H*. [1]

For Examiner's Use

(ii) The box is now submerged in water, and the beam is maintained at horizontal position.

State and explain the change, if any, to the angle that force *H* makes with the beam.

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..... [3]

[Total: 7]